



Assessment Template

Computing

Assessment:	2
Assignment Task	2
Overall marks will be awarded according to the following criteria:	4
Resources	4
General Requirements for Students:	8

Assessment:

Item	Description
Module Title:	Reinforcement Learning
Module Code:	B9AI105
Module Lecturer (s):	Dr. Anesu Nyabadza
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	Report and presentation
Percentage (%) Weighting:	40%
MIMLOs being assessed:	1, 2, 3, 4
Assessment Number:	1
Individual/Group:	Group
Issue Date:	Week 2
Submission Date:	Week 7
Feedback Date:	Week 8
Feedback Strategy:	In class and Moodle

Assignment Task**Task 1 : Report**

Solve a real-world problem using Reinforcement Learning models build from scratch as a team of 3-5. Et the idea approved by the lecturer in Week 3.

- This project requires each team (3-5 student) to identify and solve a **real-world problem using Reinforcement Learning (RL)**. The team needs to select a leader and meet at least once a week to discuss and conduct project work.
- The goal is to design, implement, and evaluate an RL solution entirely **from scratch**, demonstrating understanding of both the theoretical foundations and the practical coding elements. A final report must be submitted but it is also monitored throughout the project via a shared document on Google docs.
- Teams must clearly explain the motivation for the chosen problem, why RL is an appropriate method, and what impact the solution could have in real-world settings.
- The work must show comprehension of RL concepts such as agents, environments, states, actions, rewards, value estimation, exploration strategies, and policy optimisation.
- Teams will work collaboratively, and each member must contribute to design, coding, writing, and the presentation defence.
- The methodology should be described in detail, including:

Defining the environment, specifying state variables, action sets, and reward signals.

Justifying the chosen RL algorithm(s), such as Q-Learning, SARSA, DQN, Double DQN, DDPG, PPO, or TD3.

Designing the training pipeline, hyperparameters, update rules, and evaluation criteria.

Implementing the full codebase without copying pre-built RL libraries, except for basic numerical or plotting functions.

Running controlled experiments, logging performance, and comparing different training configurations.

Highlighting challenges such as instability, sparse rewards, or over-exploration, and describing how these were addressed.

- Results should summarise what the RL agent achieved, supported by plots, tables, and performance metrics (reward curves, convergence tests, success rates).

Explain how the agent improved over time, whether convergence was stable, and whether the solution generalised.

Discuss failed experiments, unexpected behaviours, and model limitations.

Compare at least two algorithmic or hyperparameter variants to demonstrate analytical reasoning.

- The discussion should reflect on what the results mean in the context of the real-world problem, how well the RL approach performed, and what trade-offs were observed.

Consider factors such as learning efficiency, computational cost, scalability, interpretability, and robustness.

Comment on what could be improved if more time or resources were available (e.g., better reward shaping, enhanced exploration, deeper networks, or simulated-to-real transfer).

- The conclusion should summarise the overall contribution of the project, the effectiveness of the RL solution, and what was learned about RL as a methodology for complex decision-making tasks.

State whether the project successfully addressed the original real-world problem and outline potential future directions for research or deployment.

- All references must follow **IEEE Referencing Style**, for example:

[1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018.

[2] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[3] Y. Lillicrap *et al.*, "Continuous control with deep reinforcement learning," arXiv:1509.02971, 2015.

- The final report must show consistent IEEE in-text citation and a complete reference list.

- A mandatory **presentation and defence session** will be held, where every team member must:

Present part of the work.

Defend both the technical coding aspects and the theoretical RL components.

Demonstrate deep understanding of the model, methodology, and results.

Answer questions individually to confirm ownership of the work.

- The final submission must include the full report, source code, experimental logs, and references, ensuring that the project is original, technically sound, and communicated to a professional standard.

(40 marks)

Task 2: Presentation

Defend the project through a 5-minute (timed) in-person presentation followed by a 15-minute Q&A. This presentation acts as a defence of your report and may affect the report grade if any part of the written work cannot be justified or explained.

(60 marks)

Overall marks will be awarded according to the following criteria:

Abstract:	5%
Introduction/literature review:	10%
Methodology/EDA:	20%
Results:	25%
Discussion:	30%
Conclusion:	10%

Resources

Software or tools required/ useful: Python

Templates, files, data tables provided by the lecturer: Moodle resources

Module Assessment Template

Highlight the Appropriate Selection as per the programme Module and Assessment Document

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

Grading Criteria		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)			
Scale	Grade Band	Criterion 1 - Problem formulation and justification for need for Reinforcement Learning	Criterion 2 – Clear Explanation of environment and reward system with scientific justification (experiments or literature backing)	Criterion 3 – At least 2 models or solutions compared and contrasted in a fair and scientific manner with clear conclusions	Criterion 4 – Formatting of report/presentation and team work
Exceptional	90-100%	Problem is defined with exceptional clarity and the need for Reinforcement Learning is justified with strong reasoning. Neatly publishable material.	Environment and reward system are explained with outstanding scientific depth and supported by experiments or literature. Neatly publishable material.	Two or more models are compared rigorously, with fair evaluation and insightful, well-supported conclusions. Neatly publishable material.	Report and presentation are flawlessly formatted, and teamwork is cohesive and highly professional. Neatly publishable material.
Outstanding	80-89%	Very clear problem formulation with strong justification for using RL.	Environment and reward system well described, backed by good scientific evidence.	Models are compared fairly and clearly, with well-reasoned conclusions.	High-quality formatting and presentation; teamwork clearly evident.
Excellent	70-79%	Clear problem definition with solid reasoning for applying RL.	Environment and reward system described well with reasonable literature or experimental support.	Model comparison is clear and mostly balanced, with good conclusions.	Report and presentation are well structured with only minor formatting issues; teamwork demonstrated.

Module Assessment Template

Very Good	60-69%	Mostly clear problem statement with acceptable justification for RL.	Environment and reward explanation adequate but limited in scientific backing.	Model comparison present but lacking depth or full fairness; conclusions somewhat general.	Acceptable formatting; teamwork mostly consistent.
Good	50-59%	Basic problem formulation with weak or partial RL justification.	Environment and reward system described but lacking clarity or evidence.	Minimal or shallow comparison of models; limited conclusions.	Noticeable formatting issues; teamwork inconsistent.
Sufficient	40-49%	Vague problem statement with limited reasoning for using RL.	Environment or reward system poorly described and lacking scientific basis.	Incomplete or superficial model comparison; conclusions unclear.	Several formatting problems; weak teamwork.
Somewhat/Unfulfilled	30-39%	Poorly defined problem and little to no justification for RL.	Environment and reward system unclear or inaccurate with no evidence	Model comparison missing or highly flawed; conclusions weak or absent.	Major formatting errors; teamwork not demonstrated.
Unfulfilled	0-29%	No meaningful problem formulation or reasoning for RL.	Environment and reward system not described.	No comparison of models and no valid conclusions.	Formatting unacceptable; teamwork absent.
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)					
MIMLO to which criterion is aligned:		1, 2	1, 2, 4	1, 2, 3, 4	3, 4

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
 - a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
 - a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.
 - d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.

- e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.